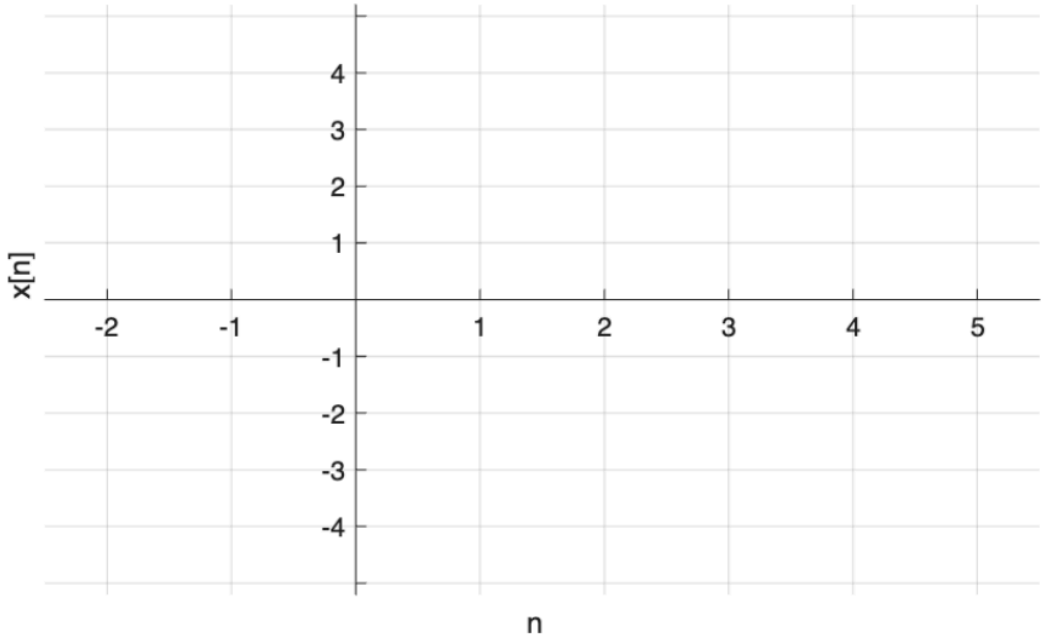
Sampling, DT signals, system properties

Practice questions · Day 02 · Sun 25 Oct · pack D1

5. Practice

From your course files

Sample Test 1 Q1 (sample test, 5 marks). Sampling.
(a) The continuous time signal $x_a(t) = 10e^{-0.01t}\cos(400\pi t)$ (t in seconds) is sampled at $F_s = 100$ Hz. Determine the sampling interval T_s and evaluate the samples of $x(n)$ for $0 \le n \le 3$. [3]
(b) On the axes provided, sketch the stem plot of the DT signal $x(n) = 2^{-n}\big(u(n+1) - u(n-2)\big) + \delta(n-3)$. [2]



Sample Test 1 Q1(b): the axes provided on the test paper

Sample Test 1 Q2 (sample test, 4 marks). DT Signal Operations. Consider the DT signal $x(n) = \{8,\ 9,\ \underline{1},\ 9,\ 6,\ 1\}$ (the underline marks $n = 0$). Samples at all other indices n are zero.
(a) Write in sequential form the signal $x(n-3) + x(n+1)$. [1]
(b) Write in sequential form the signal $x(\frac{1}{2}n)$, i.e. upsampled by a factor of 2. [1]
(c) Write in sequential form the signal $x(n)(-1)^n$. [1]
(d) Calculate the energy of $x(n)$. [1]

Sample Test 1 Q3 (sample test, 4 marks). DT System Properties. For each of the following systems, tick all the boxes that apply: ☐ Linear ☐ Time invariant ☐ Causal ☐ Static/memoryless.

(a) $y[n] - ny[n-1] = x[n] + x[n-1]$. [1]

(b) $y[n] - y[n-1] + y[n-2] = \cos(0.2\pi)x[n]$. [1]

(c) $y[n] - 0.1y[n-1] = x[n+1] - 1$. [1]

(d) $y[n] - y[n-1] - y[n-4] = x[n] + x[n+1]$. [1]

T1 Q1 (tutorial). Determine whether or not each of the following signals is periodic. In case a signal is periodic, specify its fundamental period.

(a) $x(t) = 3\cos(5t + \pi/6)$

(b) $x(n) = 3\cos(5n + \pi/6)$

(c) $x(n) = 2\exp\left(j(n/6 - \pi)\right)$

(d) $x(n) = \cos(n/8)\cos(\pi n/8)$

(e) $x(n) = \cos(\pi n/2) - \sin(\pi n/8) + 3\cos(\pi n/4 + \pi/3)$

T1 Q2 (Nyquist rate, DT frequencies and ideal D/A output for $x(t) = \sin(480\pi t) + 3\sin(720\pi t)$ sampled at 600 Hz) is worked in full as Example 1 in Section 3. Cover the working and redo it: 720 Hz; $\omega = 0.8\pi$ rad; $y(t) = -2\sin(480\pi t)$.

T1 Q3 (tutorial). The discrete time signal $x(n) = 6.35\cos(\pi n/10)$ is quantised with a resolution (a) $\Delta = 0.1$ or (b) $\Delta = 0.02$. How many bits are required in the A/D converter in each case?

T1 Q4 (tutorial). Determine the bit rate and the resolution in the sampling of a seismic signal with dynamic range of 1 Volt if the sampling rate is $F_s = 20$ samples/s and we use an 8-bit A/D converter. What is the maximum seismic signal frequency that can be recovered from the resulting digital signal without distortion?

T2 Q1 (tutorial). Determine whether or not each of the following systems is static, linear, time-invariant, causal and stable.

(a) $y(n) = \cos\big(x(n)\big)$
(b) $y(n) = \sum_{k=-\infty}^{n+1} x(k)$
(c) $y(n) = x(n)\cos(\omega_0 n)$
(d) $y(n) = x(-n+2)$
(e) $y(n) = |x(n)|$
(f) $y(n) = x(2n)$

T2 Q2 (tutorial). The following input-output pairs have been observed during the operation of a *time-invariant* system T:

$$x_1(n) = \{\underset{\uparrow}{1}, 0, 2\} \xrightarrow{T} y_1(n) = \{\underset{\uparrow}{0}, 1, 2\}$$

$$x_2(n) = \{\underset{\uparrow}{0}, 0, 3\} \xrightarrow{T} y_2(n) = \{\underset{\uparrow}{0}, 1, 0, 2\}$$

$$x_3(n) = \{\underset{\uparrow}{0}, 0, 0, 1\} \xrightarrow{T} y_3(n) = \{1, \underset{\uparrow}{2}, 1\}$$

Can you draw any conclusions regarding the *linearity* of the system?

New questions

N1 (easy) [3] $x(t) = 4\cos(300\pi t) + 2\sin(900\pi t)$ is sampled at $F_s = 500$ Hz. (a) Give the Nyquist rate. [1] (b) Find $x(n)$ with all frequencies in $(-\pi, \pi]$. [1] (c) Find the ideal DAC output $y(t)$. [1]

N2 (easy) [2] (a) Find the fundamental period of $x(n) = \cos(0.3\pi n) + \sin(0.45\pi n) + e^{j2\pi n/5}$. [1.5] (b) Is $\cos(0.3n)$ periodic? [0.5]

N3 (medium) [3] A sensor signal spans -2.5 V to $+2.5$ V and is sampled at 8 kHz. The resolution must be at most 1 mV. (a) Find the minimum number of bits. [1] (b) Find the actual Δ and the worst-case rounding error. [1] (c) Find the bit rate and the highest frequency that can be recovered. [1]

N4 (medium, chained) [4] $x(n) = \{2, -1, \underset{\uparrow}{3}, 1, 4\}$. (a) Find $x(n-2) - x(n+1)$. [1] (b) Find $x(-2n+1)$. [1] (c) Find the even part $x_e(n) = \frac{1}{2}[x(n) + x(-n)]$. [1] (d) Find E_x and say whether x is an energy or a power signal. [1]

N5 (medium) [3] Classify each system (static, linear, time-invariant, causal, stable), giving a counterexample for each property that fails. (a) $y(n) = e^{x(n)}$ (b) $y(n) = \sum_{k=n-2}^{n+2} x(k)$ (c) $y(n) = x(n^2)$

N6 (hard, chained) [5] A radar transmits $x(n) = \{\underset{\uparrow}{1}, 2, -1\}$. The received echo is $y(n) = 0.5\,x(n - D)$. Use the L03 definition $r_{yx}(l) = \sum_n y(n)x(n - l)$. (a) Find $r_{xx}(l)$ for all l. [1] (b) Take $D = 3$. Write $y(n)$ and compute $r_{yx}(l)$. [2] (c) Find $\hat{d}$ and show that $r_{yx}(l) = 0.5\,r_{xx}(l - D)$. [1] (d) Take $F_s = 1$ MHz and $c = 3 \times 10^8$ m/s. Estimate the range r. [1]

6. Quick recall check

1. Is $\cos(0.7n)$ periodic?
2. With $F_s = 1$ kHz, at what frequency does a 1.3 kHz cosine appear after ideal reconstruction?
3. How many bits does a 300-level quantiser need?
4. Is $y(n) = x(n) + 2$ linear? Is it time-invariant?
5. What is $r_{xx}(0)$ equal to?

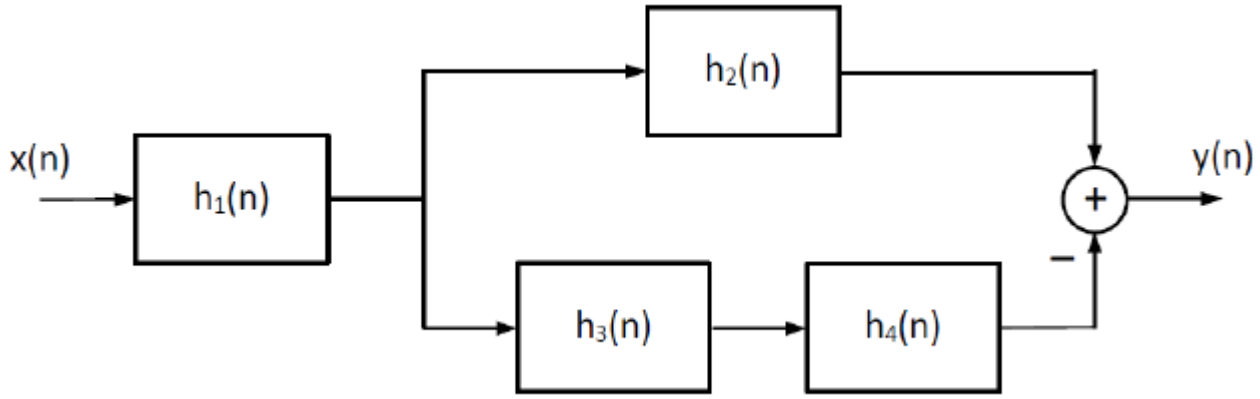
LCCDE and convolution

Practice questions · Day 04 · Tue 27 Oct · pack D2

5. Practice

From your course files

T2 Q4 (tutorial). Consider the interconnection of LTI systems shown below.
(a) Express the overall impulse response of the system $h(n)$ in terms of $h_1(n)$, $h_2(n)$, $h_3(n)$ and $h_4(n)$.
(b) Determine $h(n)$ when $h_1(n) = \{\underset{\uparrow}{2}, 1, 2\}$, $h_2(n) = h_3(n) = (n+1)u(n)$, and $h_4(n) = \delta(n-2)$. Is this system (i) causal and (ii) stable?



T2 Q4 figure (from the tutorial sheet)

T3 Q2 (tutorial). A LSI system is described by the LCCDE

$$y[n] = \frac{1}{4}\big(x[n] + 2x[n-1] + x[n-2]\big).$$

(a) Write down the characteristic modes of this system.
(b) Find the output of the system if the input signal is $x[n] = 5^{-n}u[n]$.

T3 Q4 (tutorial). A LSI system is described by the LCCDE

$$y[n] + y[n-1] + \frac{1}{4}y[n-2] = x[n].$$

The initial conditions are $y[-2] = 2$, $y[-1] = -1$.
(a) Write down the characteristic modes of this system.
(b) Find the output of the system if the input signal is $x[n] = 2^{-n}u[n]$.

T3 Q5 (tutorial). Determine the impulse response $h[n]$ of the following DT LSI systems:
(a) $y[n] = x[n] + 2x[n-1] + 3x[n-2] + 4x[n-4]$
(b) $y[n] + 0.2y[n-1] - 0.08y[n-2] = x[n]$
(c) $y[n] - 2y[n-1] + y[n-2] = x[n]$

T3 Q6 (tutorial). Compute the DT convolution $y[n] = h[n] * x[n]$ for the following signals:
(a) $x[n] = \{\underline{-1}\ \ 0\ \ 1\ \ 3\}$, $h[n] = \{-2\ \ 3\ \ \underline{1}\ \ 2\ \ 4\}$ (the underline marks $n = 0$)
(b) $x[n] = 3^n u[n]$, $h[n] = 2^{-n}u[n]$.
(c) $x[n] = \exp(j\omega_0 n)u[n]$, $h[n] = 2^{-n}u[n]$, where ω_0 is a constant.
(d) $x[n] = \cos(\omega_0 n)u[n]$, $h[n] = 2^{-n}u[n]$, where ω_0 is a constant.

New questions

Q1 (easy) [2]. $x(n) = \{1, \underset{\uparrow}{2}, -1\}$ and $h(n) = \{\underset{\uparrow}{3}, 0, 1, 2\}$. Find $y(n) = x(n) * h(n)$, marking $n = 0$.

Q2 [3]. Find $y(n) = x(n) * h(n)$ in closed form for $x(n) = u(n) - u(n-4)$ and $h(n) = (0.5)^n u(n)$.

Q3 [3]. $y(n) - 0.5y(n-1) = x(n) - x(n-1)$. Find $h(n)$ and state whether the system is causal, stable, FIR or IIR.

Q4 [4]. $y(n) - y(n-1) + 0.25y(n-2) = x(n)$, initially at rest, with $x(n) = (1/3)^n u(n)$. Find $y(n)$ in closed form.

Q5 [3]. $y(n) - y(n-1) + 0.5y(n-2) = x(n)$. (a) Find the characteristic modes. (b) Find $h(n)$ in real form. (c) Is the system stable?

Q6 (chained) [7]. System S_1 is $y(n) = 0.5y(n-1) + x(n)$. System S_2 has $h_2(n) = (0.25)^n u(n)$. They are connected in cascade.

a. Write $h_1(n)$. [1]
b. Find the overall $h(n)$ in closed form. [2]
c. Show that $h(n)$ satisfies $y(n) - 0.75y(n-1) + 0.125y(n-2) = x(n)$ by checking $h(0)$ and $h(1)$, and explain why the modes are 0.5^n and 0.25^n. [2]

d. Find the cascade's output for $x(n) = (0.5)^n u(n)$. [2]



Q6: S_1 in cascade with S_2

6. Quick recall check

1. Which coefficients form the characteristic equation, and what are the "modes" of an FIR LCCDE?
2. What homogeneous solution do you use for a double root at $\lambda = -\frac{1}{2}$?
3. Write $a^n u(n) * b^n u(n)$ for $a \neq b$, and for $a = b$.
4. Is $h(n) = \cos(\pi n/3)u(n)$ stable? Why?
5. x has 4 samples starting at $n = -3$ and h has 5 samples starting at $n = 0$. Where does y start, and how long is it?

DTFT and its properties

Practice questions · Day 06 · Thu 29 Oct · pack D3

5. Practice

From your course files

T4 Q1 (tutorial). Compute the Fourier transform of the following signals.

(a) $x(n) = u(n) - u(n-6)$

(b) $x(n) = 2^n u(-n)$

(c) $x(n) = \left(\frac{1}{3}\right)^n u(n+4)$

(d) $x(n) = \{-2, -1, \underset{\uparrow}{0}, 1, 2\}$

T4 Q2 (tutorial). Consider the signal

$$x(n) = \{-1, 2, \underset{\uparrow}{-3}, 2, -1\}$$

with Fourier transform $X(\omega)$. Compute the following quantities without explicitly computing $X(\omega)$:

(a) $X(0)$

(b) $\angle X(0)$

(c) $\int_{-\pi}^{\pi} X(\omega)\, d\omega$

(d) $X(\pi)$

(e) $\int_{-\pi}^{\pi} |X(\omega)|^2\, d\omega$

T4 Q3 (tutorial). The following input-output pairs have been observed during the operation of various LTI systems. Determine the frequency response of each system.

(a) $x(n) = \left(\frac{1}{2}\right)^n u(n) \xrightarrow{T_1} y(n) = \left(\frac{1}{8}\right)^n u(n)$

(b) $x(n) = e^{j\frac{\pi}{5}n} \xrightarrow{T_2} y(n) = 3e^{j\left(\frac{\pi}{5}n - \frac{\pi}{3}\right)}$

(c) $x(n) = u(n) - u(n-1) \xrightarrow{T_3} y(n) = \left(\frac{1}{5}\right)^{n+2} u(n+2)$

T4 Q4 (tutorial). Consider an FIR system described by the difference equation

$$y(n) = x(n) + x(n-4).$$

(a) Compute and sketch the magnitude and phase response of this system.
(b) Determine its response to the input signals:
(i) $x(n) = \cos\left(\frac{\pi}{4}n\right) + 3\sin\left(\frac{\pi}{3}n + \frac{\pi}{10}\right)$
(ii) $x(n) = 10 + 5\cos\left(\frac{2\pi}{5}n + \frac{\pi}{2}\right)$

New questions

N1 [3 marks] (easy). $x(n) = \{1, \underset{\uparrow}{1}, 1\}$.

a. Find $X(\omega)$ in real form. [2]
b. Evaluate $X(0)$ and $X(\pi)$. [1]

N2 [4 marks]. $x(n) = (-0.5)^n u(n+1)$.

a. Find $X(\omega)$. [2]
b. Find $X(0)$ and $X(\pi)$ using the shortcuts, and check them against (a). [1]
c. Is $x(n)$ a lowpass or a highpass signal? [1]

N3 [5 marks]. $x(n) = \{2,\ -1,\ \underset{\uparrow}{3},\ 0,\ 1\}$ has DTFT $X(\omega)$. Without computing $X(\omega)$, find:

a. $X(0)$ (b) $X(\pi)$ (c) $\int_{-\pi}^{\pi} X(\omega)d\omega$ (d) $\int_{-\pi}^{\pi}|X(\omega)|^2 d\omega$ (e) $\int_{-\pi}^{\pi} X(\omega)e^{j2\omega}d\omega$ (f) $\left.\frac{dX}{d\omega}\right|_{\omega=0}$ [5 total]

N4 [6 marks] (multi-part, chained). Let $x(n) = (0.5)^n u(n)$ and $y(n) = x(n) * x(n)$.

a. Find $Y(\omega)$. [1]
b. Use differentiation in frequency to show $y(n) = (n+1)(0.5)^n u(n)$. [2]
c. Find $Y(0)$ and $Y(\pi)$, and check $Y(0)$ from the time domain. [1]
d. Let $v(n) = y_{(2)}(n)$ (time expansion by 2). Find $V(\omega)$, $V(0)$ and $V(\pi)$. [2]

N5 [5 marks] (sampling). $x_a(t) = \cos(2\pi\, 150t) + 0.5\cos(2\pi\, 400t)$ is sampled at $F_s = 500$ Hz.

a. Find the digital frequencies of both tones in $(-\pi, \pi]$. [2]
b. Sketch or describe $X(\omega)$ on $(-\pi, \pi]$. [1]
c. What does an ideal lowpass reconstruction filter (gain T, cutoff 250 Hz) output? [1]
d. What is the smallest F_s that avoids aliasing? [1]

N6 [5 marks] (hard: modulation and windowing). Let $x(n) = u(n) - u(n-6)$ (the pulse in the figure) and $y(n) = x(n)\cos(\pi n)$.

a. Use the modulation property to write $Y(\omega)$ in terms of $X(\omega)$. [2]
b. Find $Y(0)$ and $Y(\pi)$. [1]
c. Find $\int_{-\pi}^{\pi}|Y(\omega)|^2 d\omega$. [1]

d. In one sentence, describe $|Y(\omega)|$ compared with the figure. [1]

6. Quick recall check

1. Write $X(\pi)$ for $x(n) = \{4, \underset{\uparrow}{1}, 2\}$.
2. $x(n) \leftrightarrow X(\omega)$. What is the DTFT of $x(-n-3)$?
3. A 1.2 kHz tone is sampled at 2 kHz. What is the digital frequency in $(-\pi, \pi]$, and what is the apparent analogue frequency?
4. For the DTFS, which side carries the $1/N$, and what does c_0 represent?
5. $x(n)$ is real and odd. What can you say about $X(\omega)$?

Frequency response

Practice questions · Day 07 · Fri 30 Oct · pack D4

5. Practice

From your course files

T4 Q3 (tutorial). The following input-output pairs have been observed during the operation of various LTI systems. Determine the frequency response of each system.

(a) $x(n) = \left(\frac{1}{2}\right)^n u(n) \xrightarrow{T_1} y(n) = \left(\frac{1}{8}\right)^n u(n)$

(b) $x(n) = e^{j\frac{\pi}{5}n} \xrightarrow{T_2} y(n) = 3e^{j\left(\frac{\pi}{5}n - \frac{\pi}{3}\right)}$

(c) $x(n) = u(n) - u(n-1) \xrightarrow{T_3} y(n) = \left(\frac{1}{5}\right)^{n+2} u(n+2)$

T4 Q4 ($y(n) = x(n) + x(n-4)$): compute and sketch the magnitude and phase response, then the responses to (i) $x(n) = \cos(\frac{\pi}{4}n) + 3\sin(\frac{\pi}{3}n + \frac{\pi}{10})$ and (ii) $x(n) = 10 + 5\cos(\frac{2\pi}{5}n + \frac{\pi}{2})$) is worked in full as Example 1, which reproduces the official solution. Cover it and redo it without looking.

T4 Q5 (tutorial). An LTI system cannot produce frequencies at its output that are different from those applied in its input. Thus, if a system creates “new” frequencies, it must be nonlinear and/or time varying. Determine the frequency content of the outputs of the following systems to the input signal

$$x(n) = A\cos\left(\frac{\pi}{4}n\right)$$

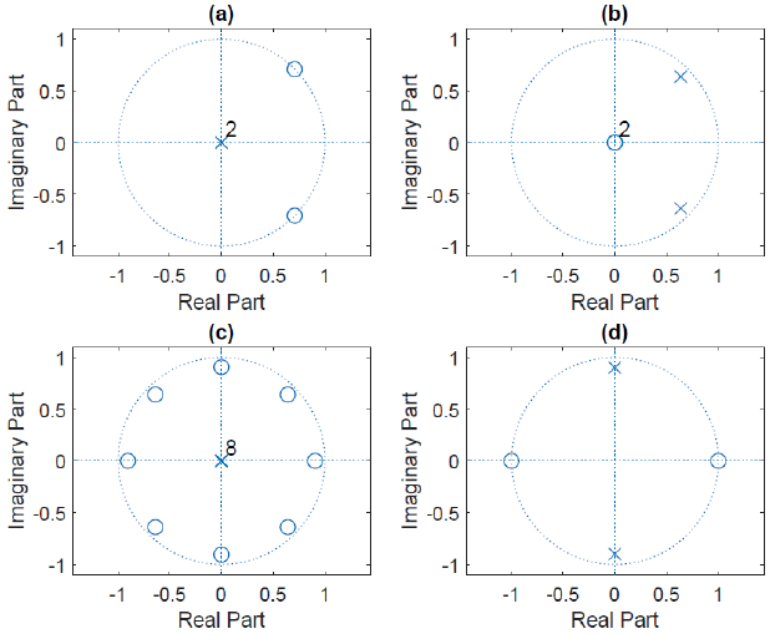
and conclude whether they are LTI.

(a) $y(n) = x(2n)$

(b) $y(n) = x^2(n)$

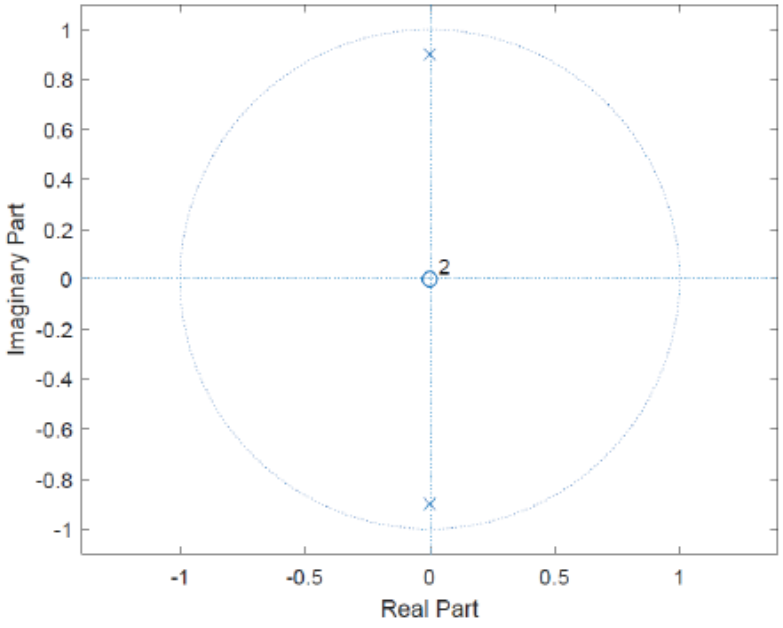
(c) $y(n) = x(n)\cos(\pi n)$

T4 Q6 (tutorial). Sketch roughly the magnitude of Fourier transforms, $|X(\omega)|$, corresponding to the pole-zero patterns given below.



T4 Q6 figure (from the tutorial sheet). o = zero, x = pole; the small numbers give multiplicity

T4 Q7 (tutorial). Sketch roughly the phase of the Fourier transform, $\theta(\omega)$, corresponding to the pole-zero pattern given below (assume $b_0 = 1$ and note that $N - M = 2$).



T4 Q7 figure (from the tutorial sheet)

New questions

N1. [3] An FIR system has $h(n) = \{\underset{\uparrow}{1}, 2, 1\}$.

a. Find $H(\omega)$ in the form $e^{-j\alpha\omega}R(\omega)$ with $R(\omega)$ real. [1]
b. Find the output for $x(n) = 4\cos(\frac{\pi}{3}n + \frac{\pi}{4})$. [1]
c. Which input frequency in $[0, \pi]$ is completely blocked? [1]

N2. [4] A causal FIR system is known to have the form $y(n) = b_0x(n) + b_1x(n-1)$. Two experiments are observed: the constant input $x(n) = 1$ gives $y(n) = 3$, and $x(n) = (-1)^n$ gives $y(n) = -(-1)^n$.

a. Find b_0 and b_1. [2]
b. Find the output for $x(n) = \cos(\frac{\pi}{2}n)$. [2]

N3. [4] Apply $x(n) = A\cos(\frac{\pi}{3}n)$ to each system. Give the output frequencies in $[0, \pi]$ and state what you can conclude about LTI.

a. $y(n) = x(n)x(n-1)$ (b) $y(n) = x(n) + x(n-2)$ (c) $y(n) = x(-n)$ (d) $y(n) = 2x(n) + 1$

N4. [9] A notch filter is described by $y(n) = -0.81y(n-2) + x(n) + x(n-2)$.

a. Find $H(\omega)$. [1]
b. Find its poles and zeros and sketch the pole-zero plot. [2]
c. Using the vector method, evaluate $|H(\pi/4)|$ and $\theta(\pi/4)$. [3]
d. Find the steady-state output for $x(n) = 5 + 2\cos(\frac{\pi}{2}n) + 3\cos(\frac{\pi}{4}n + \frac{\pi}{3})$. [2]
e. Sketch $|H(\omega)|$ for $0 \le \omega \le \pi$. [1]

N5. [8] A first-order system is $H(\omega) = b_0 \dfrac{1 + e^{-j\omega}}{1 - 0.8e^{-j\omega}}$.

a. Choose $b_0 > 0$ so that $H(0) = 1$. [1]
b. Find $h(n)$, and confirm $\sum_n h(n) = H(0)$. [2]
c. Using the vectors from the zero and the pole to $e^{j\pi/2}$, find $|H(\pi/2)|$ and $\theta(\pi/2)$. [3]
d. Find the steady-state output for $x(n) = \cos(\frac{\pi}{2}n) + (-1)^n$. [1]
e. Find the 3 dB cutoff ω_c, where $|H(\omega_c)| = 1/\sqrt{2}$. [1]

6. Quick recall check

1. What is the steady-state output of an LTI system to $x(n) = 7$ (a constant)?
2. $H(\omega_0) = -2$. Write the output for $\cos(\omega_0 n + 0.3)$.
3. A zero sits exactly on the unit circle at angle $2\pi/3$. What is $|H(2\pi/3)|$?
4. Does a pole at the origin change $|H(\omega)|$? Does it change $\theta(\omega)$?
5. An input at $\pi/5$ gives outputs at $\pi/5$ and $2\pi/5$. Could the system be LTI?

DFT and circular convolution

Practice questions · Day 09 · Sun 1 Nov · pack D5

5. Practice

From your course files

T5 Q1 (tutorial). Determine the circular convolution of the sequences

$$x_1(n) = \{\underset{\uparrow}{1}, 2, 3, 1\} \qquad x_2(n) = \{\underset{\uparrow}{4}, 3, 2, 2\}$$

using the time-domain circular convolution formula.

T5 Q2 (tutorial). Repeat the above problem using the 4-point DFT and IDFT.

T5 Q3 (tutorial). Compute the 8-point DFT of the sequence

$$x(n) = \begin{cases} 1, & 0 \le n \le 3 \\ 0, & 4 \le n \le 7 \end{cases}$$

Using the DFT obtained above, compute the DFT of the following sequences

$$x_1(n) = \begin{cases} 1, & n = 0 \\ 0, & 1 \le n \le 4 \\ 1, & 5 \le n \le 7 \end{cases} \qquad x_2(n) = \begin{cases} 0, & 0 \le n \le 1 \\ 1, & 2 \le n \le 5 \\ 0, & 6 \le n \le 7 \end{cases}$$

T5 Q6 (tutorial).

(a) Determine the discrete-time Fourier transform $X(\omega)$ of the signal $x(n) = \{1, 2, \underset{\uparrow}{3}, 2, 1, 0\}$.

(b) Compute the 6-point DFT $V(k)$ of the signal $v(n) = \{\underset{\uparrow}{3}, 2, 1, 0, 1, 2\}$.

(c) What is the relationship between $X(\omega)$ and $V(k)$? Explain.

T5 Q4 (tutorial). Consider a causal LTI system with transfer function

$$H(z) = \frac{1}{1 - 0.5z^{-1}}.$$

The output signal of the system, $y(n)$, is known for $0 \leq n \leq 63$. Assuming that $H(z)$ is also known, develop a method to recover the input signal $x(n)$ for $0 \leq n \leq 63$ using the 64-point DFT and IDFT.

New questions

Q1 (easy) [3 marks]. Compute the 4-point DFT of $x(n) = \{1, 0, 2, 1\}$ using the $\mathbf{W}_4$ matrix. Verify that $X(3) = X^*(1)$.

Q2 [4 marks]. $x(n) = \{1, 2, 3, 4, 5, 6\}$, $n = 0, \ldots, 5$.

a. Its DTFT is sampled at $\omega_k = 2\pi k/4$, $k = 0, \ldots, 3$. Find the sequence $\tilde{x}(n)$ returned by the 4-point IDFT of these samples. [2]

b. Find $\tilde{X}(0)$ and confirm it equals $X(\omega)$ at $\omega = 0$. [1]

c. What is the smallest N for which the IDFT returns $x(n)$ exactly? [1]

Q3 [5 marks]. $x_1(n) = \{1, 2, 0, 1\}$, $x_2(n) = \{1, 1, 1\}$.

a. Compute $x_1 \circledast_4 x_2$ by the matrix method. [3]
b. Compute the linear convolution and show how (a) follows from it. [2]

Q4 [6 marks, chained]. An FIR filter $h(n) = \{1, -1\}$ is driven by $x(n) = \{1, 2, 1\}$.

a. What is the minimum DFT length for the DFT method to give the linear convolution? [1]
b. Compute the 4-point DFTs $X(k)$ and $H(k)$. [2]
c. Form $Y(k) = X(k)H(k)$ and find $y(n)$ by IDFT. [2]
d. Explain why $Y(k) = Y(\omega)|_{\omega = \pi k/2}$ here, and what $y(n)$ would be with $N = 3$. [1]

Q5 [6 marks]. $x(n) = \{1, 2, 3, 4\}$ has 4-point DFT $X(k) = \{10, -2 + 2j, -2, -2 - 2j\}$. Without computing any new DFT sum, find the 4-point DFT of:

a. $x_a(n) = x((n-1))_4$ [2]
b. $x_b(n) = x((-n))_4$ [2]
c. $x_c(n) = (-1)^n x(n)$ [1]
d. Use Parseval to find $\sum_n x^2(n)$ from $X(k)$. [1]

Q6 (hard) [6 marks]. A sequence $x(n)$ of length 3 was passed through $h(n) = \{2, 1\}$, giving $y(n) = \{2, 5, 6, 2\}$.

a. Recover $x(n)$ using 4-point DFTs. [4]
b. Explain why the same method would fail if $h(n) = \{1, 1\}$ and $N = 4$. [2]

6. Quick recall check

1. What is $((-3))_5$?
2. Write the second row of $\mathbf{W}_4$.
3. Minimum DFT length to linearly convolve sequences of length 10 and 5?
4. Real $x(n)$, $N = 8$, $X(3) = 1 + 2j$. What is $X(5)$?
5. Direct DFT vs radix-2 FFT complex multiplications for $N = 256$?

Spectral analysis and block convolution

Practice questions · Day 11 · Tue 3 Nov · pack D6

5. Practice

From your course files

T5 Q5 (tutorial). Let $x_a(t)$ be an analogue signal with bandwidth $B = 3$ kHz. We wish to use a $N = 2^m$-point DFT to compute the spectrum of the signal with a resolution less than or equal to 50 Hz. Determine
(a) the minimum sampling rate,
(b) the minimum number of required samples
(c) the minimum length of the analogue signal record.

T5 Q4 (tutorial). Consider a causal LTI system with transfer function

$$H(z) = \frac{1}{1 - 0.5z^{-1}}.$$

The output signal of the system, $y(n)$, is known for $0 \leq n \leq 63$. Assuming that $H(z)$ is also known, develop a method to recover the input signal $x(n)$ for $0 \leq n \leq 63$ using the 64-point DFT and IDFT.

T5 Q6 (tutorial).
(a) Determine the discrete-time Fourier transform $X(\omega)$ of the signal $x(n) = \{1, 2, \underset{\uparrow}{3}, 2, 1, 0\}$.
(b) Compute the 6-point DFT $V(k)$ of the signal $v(n) = \{\underset{\uparrow}{3}, 2, 1, 0, 1, 2\}$.
(c) What is the relationship between $X(\omega)$ and $V(k)$? Explain.

New questions

N1. [3 marks] A signal is sampled at $F_s = 1$ kHz and analysed with a $N = 200$ point DFT (no zero-padding).

a. What is the frequency resolution and the record length? [1]
b. A real tone at 120 Hz is present. In which bins does it appear? [1]
c. What analogue frequency does bin $k = 37$ represent? What about $k = 150$? [1]

N2. [4 marks] $L = 100$ samples of a signal are taken at $F_s = 1$ kHz and zero-padded to $N = 512$.

a. Find the bin spacing and the frequency resolution. [2]
b. The signal contains tones at 200 Hz and 205 Hz. Can this DFT separate them? Would zero-padding to 4096 help? [1]
c. What is the minimum number of samples needed to separate them, and the matching power-of-2 DFT size without padding? [1]

N3. [5 marks] An FIR filter $h(n) = \{\underset{\uparrow}{1}, -1\}$ is applied to $x(n) = \{\underset{\uparrow}{1}, 0, 2, 1, 3, 1\}$ using the overlap-add method with blocks of $L = 3$ input samples.

a. What DFT length N is needed? Write out the padded $h(n)$. [1]
b. Find the output of each block. [2]
c. Combine them to get $y(n)$ and check one value directly. [2]

N4. [8 marks] The filter $h(n) = \{\underset{\uparrow}{2}, -1, 1\}$ is applied to $x(n) = \{\underset{\uparrow}{1}, 3, 2, 1, 0, 2, 4, 1\}$ using the overlap-save method with $N = 5$ point DFTs.

a. How many new input samples does each block carry? [1]
b. Write out every input block needed to produce the complete output. How many blocks are there? [2]
c. Compute the 5-point circular convolution of h with the first two blocks. Say which outputs you discard and why. [3]
d. Give the complete output $y(n)$. How many blocks would overlap-add have needed with the same N? [2]

N5. [5 marks] A signal sampled at $F_s = 8$ kHz contains a strong tone at 1010 Hz (amplitude 1) and a weak tone at 1050 Hz (amplitude 0.01, i.e. -40 dB). You are told that the worst-case leakage of a tone into a bin d or more bins away is:

d (bins)	2	3	4	5	6
Rectangular	-18 dB	-21 dB	-23 dB	-25 dB	-26 dB
Hanning	-31 dB	-41 dB	-48 dB	-54 dB	-58 dB

a. Which window must be used, and why? [2]
b. Find the minimum power-of-2 record length L (with $N = L$) so that the weak tone is at least 5 bins from the strong one. Give T_0. [2]
c. What would you see with a rectangular window and the same L? [1]

N6. [9 marks] An analogue signal band-limited to 5 kHz is sampled by an ADC fixed at $F_s = 12$ kHz. It contains two equal-amplitude tones at 1500 Hz and 1530 Hz.

a. Is F_s adequate? [1]

b. Using a rectangular window and $N = L = 2^m$, find the minimum N that resolves the tones, the record length and the actual resolution. [2]
c. For that N, find the bin index of each tone. Comment on leakage. [2]
d. A colleague records only 128 samples and zero-pads them to 2048. Find the bin spacing and the resolution. Will the tones appear as two peaks? [2]
e. The colleague then insists on a Hanning window. Using the rule of thumb that Hanning needs about twice the record length, find the new minimum power-of-2 L and T_0. [2]

6. Quick recall check

1. $F_s = 2$ kHz, $N = 256$, no zero-padding. What is the resolution, and what is T_0?
2. Does zero-padding from 256 to 1024 samples let you resolve tones that were unresolved before? What does it do?
3. Write the 5-point Hanning window from the slide 32 formula.
4. Overlap-save, $M = 4$, $N = 16$: how many new samples per block, and which outputs do you discard?
5. Give the main-lobe and side-lobe trade-off of the Hanning window compared with the rectangular one, in one sentence.

DSP · L07

z-transform and ROC

Practice questions · Day 13 · Thu 5 Nov · pack D7

5. Practice

From your course files

T6 Q1 (tutorial). Determine the z-transforms of the following signals and sketch their ROC.
(a) $x_1(n) = \left(\frac{1}{3}\right)^n u(n) + 2^n u(-n-1)$
(b) $x_2(n) = \left(\left(\frac{1}{3}\right)^n - 2^n\right) u(n)$
(c) $x_3(n) = x_1(n+4)$
(d) $x_4(n) = x_1(-n)$

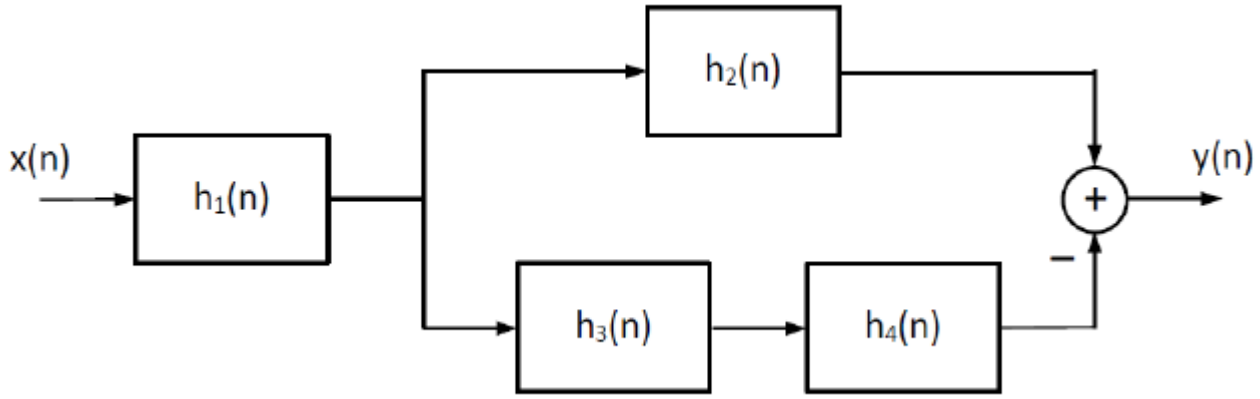
T6 Q2 (tutorial). Compute the convolution of the following signals using the z-transform

$$x_1(n) = \left(\frac{1}{3}\right)^n u(n) + 2^n u(-n-1), \qquad x_2(n) = \left(\frac{1}{2}\right)^n u(n).$$

T6 Q3 (tutorial). Determine the transfer function and impulse response of the system shown below where

$$h_1(n) = \left\{\frac{1}{2}, \underset{\uparrow}{\frac{1}{4}}, \frac{1}{2}\right\}, \qquad h_2(n) = h_3(n) = \{1, \underset{\uparrow}{-1}\}, \qquad h_4(n) = \delta(n+1) + 2\delta(n-2).$$

Is this system causal?



T6 Q3 figure (from the tutorial sheet)

New questions

N1. [3 marks] $x(n) = \{\underset{\uparrow}{1}, -2, 1\}$. (a) Find $X(z)$ and its ROC. (b) Find all poles and zeros. (c) Give the transform and ROC of $x(n+2)$.

N2. [4 marks] Find the z-transform, ROC and pole-zero locations of $x(n) = \left(\frac{1}{2}\right)^n \cos\left(\frac{\pi n}{2}\right) u(n)$.

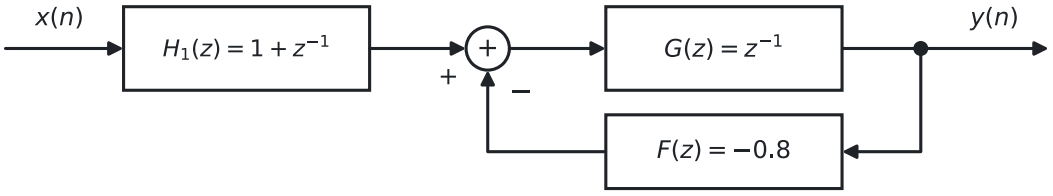
N3. [6 marks] $x(n) = \left(\frac{1}{2}\right)^n u(n) \leftrightarrow X(z) = \dfrac{1}{1 - \frac{1}{2}z^{-1}}$, $|z| > \frac{1}{2}$. Using properties only, give the transform and ROC of: (a) $x(n-3)$; (b) $x(-n)$; (c) $2^n x(n)$; (d) $n\,x(n)$; (e) $x_{(2)}(n)$; (f) find $x(0)$ from $X(z)$.

N4. [5 marks] $x(n) = \left(\frac{1}{4}\right)^n u(n)$ is applied to an LTI system with $h(n) = \delta(n+1) - \frac{1}{4}\delta(n)$.

a. Find $X(z)$ and $H(z)$ with their ROCs. (b) Find $Y(z)$, its ROC, and $y(n)$. (c) Is the system causal? Explain why the ROC of $Y(z)$ is larger than $R_x \cap R_h$.

N5. [6 marks] A system has $h_1(n)$ in cascade with a negative-feedback loop. The loop has forward path $G(z) = z^{-1}$ and feedback path $F(z) = -0.8$ (so the net loop is positive feedback with gain 0.8). $h_1(n) = \{\underset{\uparrow}{1}, 1\}$.

a. Find the loop transfer function $H_L(z)$ and the overall $H(z)$. (b) List the poles and zeros of $H(z)$, including any at 0 or ∞. (c) Is the system causal? Give the ROC. (d) Does $H(\omega)$ exist? If yes, find $|H(\omega)|$ at $\omega = 0$ and $\omega = \pi$.



N5: $h_1(n)$ in cascade with the feedback loop

N6. [6 marks] $x(n) = a^{|n|}$ with a real.

a. Find $X(z)$ and the ROC, and state the condition on a for the transform to exist. (b) For $a = \frac{1}{2}$, give $X(z)$, the poles and zeros, and sketch the ROC. (c) For $a = \frac{1}{2}$, find $X(\omega)$ at $\omega = 0$ and $\omega = \pi$. (d) What happens for $a = 2$?

6. Quick recall check

1. What is the ROC shape of a causal exponential, and which pole sets its boundary?
2. $Z\{x(n+3)\} = ?$ in terms of $X(z)$.
3. If $X(z)$ has ROC $0.2 < |z| < 4$, what is the ROC of $Z\{x(-n)\}$?
4. Where is the DTFT found in the z-plane, and when does it exist?
5. $H(z) = z + 2z^{-2}$. Is the system causal? Why?

Inverse z-transform

Practice questions · Day 15 · Sat 7 Nov · pack D8

5. Practice

From your course files

T7 Q1 (tutorial). Determine all possible signals $x(n)$ with the following z-transform:

$$X(z) = \frac{5z^{-1}}{(1-2z^{-1})(3-z^{-1})}$$

T7 Q2 (tutorial). Determine the signal $x(n)$ with the z-transform:

$$X(z) = \frac{3}{1-\frac{10}{3}z^{-1}+z^{-2}}$$

if $x(n)$ is absolutely summable.

T7 Q3 (tutorial). A *causal* discrete-time LTI system produces the output

$$y(n) = \left(\frac{1}{3}\right)^n u(n)$$

in response to the input signal

$$x(n) = \left(\frac{1}{2}\right)^n u(n) - \frac{1}{4}\left(\frac{1}{2}\right)^{n-1} u(n-1).$$

(a) Determine the impulse response $h(n)$ and transfer function $H(z)$ of this system.
(b) Find the difference equation that characterises the system.
(c) Is this system stable?

T8 Q5(c) (tutorial, optional redo). A communication channel is modelled by the transfer function

$$H(z) = \frac{1 + 0.7z^{-1} - 1.2z^{-2}}{1 - 0.8z^{-1} + 0.64z^{-2}}, \qquad \text{ROC: } |z| > 0.8$$

(c) Determine the transfer function and impulse response of its stable inverse. Is it causal, anticausal or noncausal?

New questions

Q1. Power series [4 marks]

a. $X(z) = 2z^2 - z + 3 - 4z^{-2}$, ROC $0 < |z| < \infty$. Write $x(n)$ as a sequence with the $n = 0$ sample marked. [1]

b. $X(z) = \dfrac{1 + 2z^{-1}}{1 - z^{-1} + \frac{1}{2}z^{-2}}$. Use long division to find $x(0), \ldots, x(3)$ if $x(n)$ is causal. [2]

c. Same $X(z)$, but now the ROC is $|z| < 1/\sqrt{2}$. Find $x(-1)$ and $x(-2)$. [1]

Q2. LCCDE, improper $H(z)$ and system types [7 marks]

Three causal systems:

- S1: $y(n) - 0.5y(n-1) = x(n) + x(n-1)$
- S2: $y(n) = x(n) - \frac{1}{8}x(n-3)$
- S3: $y(n) = 0.9y(n-2) + x(n)$

a. Find $H_1(z)$ and its poles and zeros. [2]

b. Find $h_1(n)$ in closed form. [3]

c. Classify each system as FIR (all-zero), all-pole or pole-zero. Give its poles, and say whether it is stable. [2]

Q3. Find all possible sequences [7 marks]

$$X(z) = \frac{1 + z^{-2}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})(1 - 3z^{-1})}$$

a. Expand $X(z)$ in partial fractions. [3]

b. List every possible $x(n)$ with its ROC, and label each causal, anti-causal or two-sided. [3]

c. Which one has a DTFT? Check your answer by evaluating $X(1)$ two ways. [1]

Q4. System ID → LCCDE → stability [10 marks]

A causal LTI system has input $x(n) = u(n)$ and output

$$y(n) = \left[\frac{2}{3} - \frac{1}{5}\left(\frac{1}{2}\right)^n + \frac{8}{15}(-2)^n\right]u(n)$$

a. Find $X(z)$ and $Y(z)$ as single fractions, with ROCs. [2]

b. Find $H(z)$ and the difference equation. [3]

c. Find $h(n)$. Is the system stable? [2]

d. Another system obeys the same difference equation but is known to be stable. Find its $h(n)$. Is it causal? [2]

e. Would the system in (d) produce the same $y(n)$ for the input $u(n)$? Give a one-line reason. [1]

Q5. Improper $H(z)$ with complex poles [8 marks]

$$H(z) = \frac{1 + z^{-2}}{1 - z^{-1} + \frac{1}{2}z^{-2}}, \qquad \text{causal}$$

a. Find the poles and zeros. Is the system stable? [2]

b. Use long division to find $h(0), \ldots, h(3)$. [2]

c. Find $h(n)$ in real closed form. Check it against (b). [3]

d. What is the steady-state output for $x(n) = \cos(\pi n/2)$? [1]

6. Quick recall check

1. Invert $\dfrac{4}{1 - 0.8z^{-1}}$ with ROC $|z| < 0.8$.
2. A causal system has poles at 0.9 and -1.1. Is it stable?
3. $X(z)$ has poles at 0.5, 2 and 5. How many possible $x(n)$ are there, and which ROC gives the absolutely summable one?
4. Write the LCCDE for $H(z) = \dfrac{1 + 2z^{-1}}{1 - 0.25z^{-2}}$.
5. $H(z) = 1 - 2z^{-1} + z^{-3}$. What type of system is it, where are its poles, and is it stable?

Unilateral z-transform and mixed practice

Practice questions · Day 16 · Sun 8 Nov · pack D9

5. Practice

From your course files

T3 Q3 (tutorial; do (b), redone with $Y = HX$). A LSI system is described by the LCCDE

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = x[n] - x[n-1].$$

The system is initially at rest, i.e. $y[n] = 0$ for $n < 0$.

(a) Write down the characteristic modes of this system.
(b) Find the output of the system if the input signal is $x[n] = 3^{-n}u[n]$.

T3 Q5 (tutorial; do (b) and (c) via $H(z)$). Determine the impulse response $h[n]$ of the following DT LSI systems:

(b) $y[n] + 0.2y[n-1] - 0.08y[n-2] = x[n]$
(c) $y[n] - 2y[n-1] + y[n-2] = x[n]$

T7 Q1 (tutorial). Determine all possible signals $x(n)$ with the following z-transform:

$$X(z) = \frac{5z^{-1}}{(1-2z^{-1})(3-z^{-1})}$$

T7 Q2 (tutorial). Determine the signal $x(n)$ with the z-transform:

$$X(z) = \frac{3}{1 - \frac{10}{3}z^{-1} + z^{-2}}$$

if $x(n)$ is absolutely summable.

T7 Q3 (tutorial). A *causal* discrete-time LTI system produces the output

$$y(n) = \left(\frac{1}{3}\right)^n u(n)$$

in response to the input signal

$$x(n) = \left(\frac{1}{2}\right)^n u(n) - \frac{1}{4}\left(\frac{1}{2}\right)^{n-1} u(n-1).$$

(a) Determine the impulse response $h(n)$ and transfer function $H(z)$ of this system.
(b) Find the difference equation that characterises the system.
(c) Is this system stable?

New questions

Part A: unilateral z-transform (graded easy to hard)

Q1 [4 marks]. $y(n) - 0.5y(n-1) = x(n)$ with $x(n) = u(n)$ and $y(-1) = 4$.

a. Find $Y(z)$. [2] (b) Find $y(n)$. [1] (c) Give $y_{zi}(n)$ and $y_{zs}(n)$. [1]

Q2 [5 marks]. $y(n) - 0.7y(n-1) + 0.1y(n-2) = x(n)$ with $y(-1) = 1$, $y(-2) = 5$ and $x(n) = \delta(n)$.

a. Find the IC polynomial $I(z)$. [1] (b) Find $y_{zi}(n)$. [1] (c) Find $y_{zs}(n)$ and say why it equals $h(n)$. [2] (d) Find $y(n)$ and check $y(0)$ and $y(1)$. [1]

Q3 [5 marks]. $y(n) - 0.25y(n-2) = x(n) + x(n-1)$ with $x(n) = (\frac{1}{3})^n u(n)$, $y(-1) = 2$ and $y(-2) = -4$. Find $y_{zi}(n)$, $y_{zs}(n)$ and $y(n)$.

Q4 [6 marks]. $y(n) - y(n-1) + 0.25y(n-2) = x(n)$ with $x(n) = u(n)$, $y(-1) = 2$ and $y(-2) = 0$.

a. Find $y_{zi}(n)$. [2] (b) Find $y_{zs}(n)$. [2] (c) Find $y(n)$ and explain what happened to the $n(0.5)^n$ terms. [1] (d) Check $y(0)$ and $y(1)$ by recursion. [1]

Q5 [6 marks]. $y(n) - 0.9y(n-1) + 0.2y(n-2) = x(n)$ with $x(n) = u(n)$, $y(-2) = 0$ and $y(-1) = c$.

a. Find $H(z)$, its poles and the steady-state output. [2] (b) Find $Y(z)$ in terms of c. [2] (c) Find c so that the $(0.5)^n$ mode does not appear in $y(n)$. [1] (d) Find $y(n)$ for that c. [1]

Q6 [8 marks] (chained). A causal system is described by $y(n) - 0.25y(n-1) - 0.125y(n-2) = x(n) - x(n-1)$. The input is $x(n) = (0.5)^n u(n)$ and the ICs are $y(-1) = 4$, $y(-2) = -8$.

a. Find $H(z)$, its poles and zeros, and state whether the system is stable. [2]
b. Find $y_{zi}(n)$. [2]
c. Find $y_{zs}(n)$. Explain why a term in $n(0.5)^n$ appears. [3]
d. Give $y(n)$ and check $y(0)$ and $y(1)$ by recursion. [1]

Part B: mixed z-domain set (exam-style, about 12 min each)

Q7 [8 marks] ROC and inverse. Let $X(z) = \dfrac{2 - 2.5z^{-1}}{(1 - 0.5z^{-1})(1 + 2z^{-1})}$.

a. Find the poles and zeros, including any at $z = 0$. [1]
b. Expand $X(z)$ in partial fractions. [2]
c. List every possible ROC and the matching $x(n)$, and say whether each is causal, anti-causal or two-sided. [3]
d. Which $x(n)$ has a DTFT? Evaluate that DTFT at $\omega = 0$. [1]
e. Find the unilateral transform $X^+(z)$ of the signal in (d). Why is it not equal to $X(z)$? [1]

Q8 [8 marks] Transfer function, stability and the response with ICs. A causal system is described by $y(n) + 0.1y(n-1) - 0.3y(n-2) = x(n)$.

a. Find $H(z)$, its poles and $h(n)$. [3]
b. Is the system stable? Give a reason. [1]
c. With $x(n) = u(n)$, $y(-1) = 1$ and $y(-2) = 0$, find $y_{zi}(n)$, $y_{zs}(n)$ and $y(n)$. [3]

d. What value does $y(n)$ settle to, and why do the ICs not affect it? [1]

Q9 [8 marks] Identification, then ICs. A causal LTI system, initially at rest, gives the output $y(n) = \left[4 - 4(0.5)^n + (0.25)^n\right]u(n)$ for the input $x(n) = u(n)$.

a. Find $H(z)$. [2] (b) Find the LCCDE. [1] (c) Find $h(n)$ and state whether the system is stable. [2]
b. The same system now has $y(-1) = y(-2) = 4$, as if its output had already settled at 4. It gets the same step $u(n)$ from $n = 0$. Find $y(n)$ and explain why it is not constant. [3]

Q10 [9 marks] Unstable poles, ROC and why ICs cannot fix them. Let

$$H(z) = \frac{1.5z^{-1}}{(1 - 0.5z^{-1})(1 - 2z^{-1})}.$$

a. Find the poles and zeros, including those at 0 and ∞. [1]
b. For each possible ROC, give $h(n)$ and state whether the system is causal and whether it is stable. [3]
c. Write the LCCDE of the causal system. [1]
d. Find the step response of the causal system at rest. [1]
e. Take $y(-2) = 0$. Find the $y(-1)$ that makes the step response bounded, and give $y(n)$. [2]
f. Explain in one or two sentences why (e) is not a way to build a stable system. [1]

6. Quick recall check

1. Write $\mathcal{Z}^+\{y(n-2)\}$ in terms of $Y(z)$ and the ICs.
2. The input is $x(n) = u(n)$. What is $\mathcal{Z}^+\{x(n-1)\}$?
3. Which part of the total response equals $H(z)X(z)$, and which poles can appear in the other part?
4. Which time sequence has the transform $\frac{1}{(1-pz^{-1})^2}$?
5. $Y(z) = \frac{3 - z^{-1} + 6z^{-2}}{(1-z^{-1})(1-2z^{-1})(1+3z^{-1})}$. What is $y(0)$, with no partial fractions?

Digital filters

Practice questions · Day 17 · Mon 9 Nov · pack D10

5. Practice

From your course files

T8 Q1 (tutorial). Determine the coefficients of an FIR filter described by the difference equation

$$y(n) = b_0 x(n) + b_1 x(n-1) + b_2 x(n-2)$$

such that

i. It rejects the frequency component $\omega_0 = 2\pi/3$ completely,
ii. Its frequency response is normalised to $H(\omega) = 1$ at $\omega = 0$, and
iii. $b_0 = b_2$ so that it has a linear phase response.

Compute and sketch the magnitude and phase response of the filter to confirm that it satisfies the above requirements.

T8 Q2 (tutorial). Compute the magnitude and phase response of the comb filter

$$H(z) = \frac{1}{5} \sum_{k=0}^{4} z^{-k}$$

and determine the frequencies of continuous-time sinusoids that cannot pass through this filter if an ADC with sampling frequency $F_s = 1$ kHz is employed.

T8 Q3 (tutorial). A digital notch filter is required to remove an undesirable 50-Hz hum associated with a power supply in an ECG recording application. The sampling frequency used is 500 Hz. Design a second-order FIR notch filter for this purpose with frequency response satisfying $H(\omega) = 1$ at $\omega = \pi$.

T8 Q4 (tutorial). Consider the FIR system:

$$y(n) = x(n) - 0.9x(n-4)$$

(a) Sketch its pole-zero plot.
(b) Sketch roughly its magnitude response using the pole-zero plot.
(c) Determine the transfer function of its causal inverse system. Is it stable?
(d) Sketch roughly the magnitude response of the inverse system using the pole-zero plot.

T8 Q5 (tutorial). A communication channel is modelled by the transfer function

$$H(z) = \frac{1 + 0.7z^{-1} - 1.2z^{-2}}{1 - 0.8z^{-1} + 0.64z^{-2}}, \qquad \text{ROC: } |z| > 0.8$$

(a) Sketch its pole-zero plot.
(b) Is the channel a minimum-phase, maximum-phase or nonminimum-phase system?
(c) Determine the transfer function and impulse response of its stable inverse. Is it causal, anticausal or noncausal?

New questions

N1. [3 marks] A lowpass FIR filter has $h_{LP}(n) = \{0.25, \underset{\uparrow}{0.5}, 0.25\}$.

a. Find $h_{HP}(n) = (-1)^n h_{LP}(n)$ and $H_{HP}(z)$. [1]
b. Give the zeros of both filters and $|H|$ at $\omega = 0, \pi/2, \pi$ for both. [2]

N2. [6 marks] An instrument samples at $F_s = 1$ kHz and picks up 50 Hz mains hum.

a. Design a second-order FIR notch with unity gain at DC. [2]
b. Evaluate its gain at 100 Hz and at 500 Hz. Comment. [1]
c. Add poles at $0.95e^{\pm j\omega_0}$. Find b_0 for unity DC gain and write $H(z)$. [2]
d. Evaluate the gain of (c) at 100 Hz and 500 Hz. [1]

N3. [4 marks] A moving-average filter $y(n) = \frac{1}{M+1}\sum_{k=0}^{M} x(n-k)$ runs at $F_s = 8$ kHz.

a. Choose M so the filter nulls 1 kHz and all its harmonics up to $F_s/2$. List the null frequencies. [2]
b. Find the gain at 500 Hz and the group delay in ms. [2]

N4. [6 marks] A digital resonator at $F_s = 8$ kHz has poles at $0.95e^{\pm j\omega_0}$ with $F_0 = 1$ kHz and zeros at $z = \pm 1$.

a. Write $H(z)$ and the difference equation in terms of b_0. [2]
b. Find b_0 so that $|H(\omega_0)| = 1$. What are $|H(0)|$ and $|H(\pi)|$? [2]
c. Move the poles onto the unit circle and set $b_0 = \sin\omega_0$. Write the difference equation and the first 4 samples of the impulse response. [2]

N5. [7 marks] A channel is $H(z) = \dfrac{1 - 1.5z^{-1} - z^{-2}}{1 - 0.4z^{-1}}$, ROC $|z| > 0.4$.

a. Find the poles and zeros and classify the phase type. [2]
b. Is the causal inverse stable? [1]

c. Find the stable inverse $H_I(z)$, its ROC and $h_I(n)$. Is it causal? [3]
d. Evaluate $h_I(n)$ for $n = -2, \ldots, 2$. [1]

N6. [5 marks] $H(z) = \dfrac{0.5 + z^{-1}}{1 + 0.5z^{-1}}$ (causal).

a. Find the pole and zero. Show $|H(\omega)| = 1$. [2]
b. Is it minimum phase? Does it have a stable causal inverse? [1]
c. Show that the group delay is $\tau(\omega) = \dfrac{1 - a^2}{1 + 2a\cos\omega + a^2}$ with $a = 0.5$, and evaluate it at $\omega = 0, \pi/2, \pi$. [2]

---

6. Quick recall check

1. An FIR filter has $h(n) = \{2, -1, 3, -1, 2\}$. Is it linear phase, and what is its group delay?
2. A 10-tap moving average runs at $F_s = 2$ kHz. What is its lowest null frequency?
3. Convert $H_{LP}(z) = \dfrac{1 + 2z^{-1} + z^{-2}}{1 - 0.5z^{-1} + 0.2z^{-2}}$ to highpass.
4. A causal stable H has zeros at 0.5 and -1.2. What kind of system is its stable inverse?
5. Where are the poles of a digital oscillator, and why is it not BIBO stable?

DSP · L10

IIR design: bilinear transform and Butterworth

Practice questions · Day 19 · Wed 11 Nov · pack D11

5. Practice

From your course files

T9 Q4 (tutorial). Given the following digital filter specifications, obtain the corresponding analogue filter specifications for bilinear transformation:

$$0.8 \leq |H(\omega)| \leq 1, \quad 0 \leq \omega \leq 0.3\pi$$

$$|H(\omega)| \leq 0.1, \quad 0.6\pi \leq \omega \leq \pi.$$

T9 Q2 (tutorial). Convert the analogue filter

$$H(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9}$$

into a digital IIR filter using the bilinear transformation.

T9 Q3 (tutorial). An ideal analogue integrator is described by $H_a(s) = 1/s$.

(a) Convert the analogue integrator into a digital integrator using the bilinear transformation. Write the difference equation for the digital integrator.
(b) Roughly sketch the magnitude and phase response of the analogue integrator.
(c) Obtain the frequency response of the digital integrator $H(\omega)$, and roughly sketch the magnitude and phase responses.
(d) Compare the magnitude and phase responses in parts (b) and (c), and comment on how well they match.
(e) Considering that the digital integrator has a pole at $z = 1$, what restrictions may need to be placed on the input signal?

T9 Q6 (tutorial). The poles of an analogue Butterworth filter are located on a circle of radius 50 in the s-plane. Find the filter order N if the minimum attenuation of the filter is 80 dB at 100 Hz.

T9 Q5 (tutorial). A digital lowpass filter is required to have the following specifications:

Passband ripple: ≤ 1 dB
Stopband attenuation: ≥ 20 dB
Passband edge frequency: 3 kHz
Stopband edge frequency: 9 kHz
Sampling frequency: 30 kHz

Obtain the transfer function and difference equation for the digital filter by applying the bilinear transformation to an analogue Butterworth filter.

New questions

Question 1 (easy) [4 marks] A digital lowpass filter must satisfy $0.85 \leq |H(\omega)| \leq 1$ for $0 \leq \omega \leq 0.2\pi$ and $|H(\omega)| \leq 0.15$ for $0.5\pi \leq \omega \leq \pi$. It is to be designed with the bilinear transform ($T = 1$) from an analogue Butterworth filter.

a. Write down the analogue specifications. [2]
b. Find the minimum order N and the range of Ω_c that meets both specs. [2]

Question 2 (easy–medium) [5 marks] A first-order digital lowpass filter with its 3-dB cutoff at 1 kHz is required. $F_s = 8$ kHz. Use the analogue prototype $H_a(s) = \dfrac{\Omega_c}{s + \Omega_c}$ and the bilinear transform with $T = 1/F_s$.

a. Find the prewarped Ω_c in rad/s. [1]
b. Find $H(z)$ and the difference equation. [3]
c. A classmate skips prewarping and uses $\Omega_c = 2\pi(1000)$ rad/s. At what frequency (Hz) is their digital filter 3 dB down? [1]

Question 3 (medium) [5 marks] An analogue 4th-order Butterworth lowpass filter has its poles on a circle of radius $2\pi(500)$ rad/s.

a. Give the four poles of $H(s)$. [2]
b. Write $H(s)$ as a product of two real quadratics. [1]
c. Find the attenuation at 1.5 kHz. [1]
d. What minimum order would give at least 60 dB attenuation at 1.5 kHz with the same pole radius? [1]

Question 4 (medium–hard) [8 marks] — full design A, stopband met exactly A digital lowpass filter is required with passband ripple ≤ 1 dB up to 1 kHz, stopband attenuation ≥ 20 dB from 3 kHz, and $F_s = 8$ kHz. Design it by applying the bilinear transform ($T = 1$) to an analogue Butterworth filter, choosing Ω_c to meet the **stopband** exactly (as in T9 Q5).

a. Find the prewarped edges and the order N. [2]
b. Find Ω_c and $H(s)$. [2]
c. Find $H(z)$ and the difference equation. [3]
d. Check the design against the specs. [1]

Question 5 (hard, multi-part) [10 marks] — full design B, passband met exactly A digital lowpass filter is required with passband ripple ≤ 1 dB up to 2 kHz, stopband attenuation ≥ 20 dB from 4 kHz, and $F_s = 12$ kHz. Use the bilinear transform on an analogue Butterworth filter, with Ω_c chosen to meet the **passband** exactly (L10 slide 17).

a. Find ω_p, ω_s, the prewarped edges ($T = 1$) and N. [2]
b. Find Ω_c, the LHP poles and $H(s)$. [2]
c. Find $H(z)$ and the difference equation. [3]
d. Verify the specs and the stability. [2]
e. Show that using $T = 1/F_s$ instead of $T = 1$ gives the same $H(z)$. At what frequency in Hz is the digital filter 3 dB down? [1]

Question 6 (hard, trap-finder) [4 marks] For the specification in Question 5, a student skips prewarping. They use $\Omega_p = \omega_p = \pi/3$ and $\Omega_s = \omega_s = 2\pi/3$ directly ($T = 1$), then apply the bilinear transform.

a. What order does their calculation give? [1]
b. Suppose they keep $N = 3$ and set Ω_c to meet their (unwarped) passband exactly. Find the actual digital gain at ω_p and ω_s. Does the design meet the specs? [3]

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6. Quick recall check

1. Under the bilinear transform, where do $s = 0$ and $s = \infty$ go in the z-plane?
2. Write the prewarping formula, and say which spec quantities it changes.
3. A passband ripple of 1 dB means $\delta_p = ?$ In the order formula, what is $(1 - \delta_p^2)/\delta_p^2$?
4. The Butterworth poles of $H(s)$ lie on what circle, and at what angles for $N = 3$?
5. The order formula gives $N = 3.02$. What order do you use, and why?

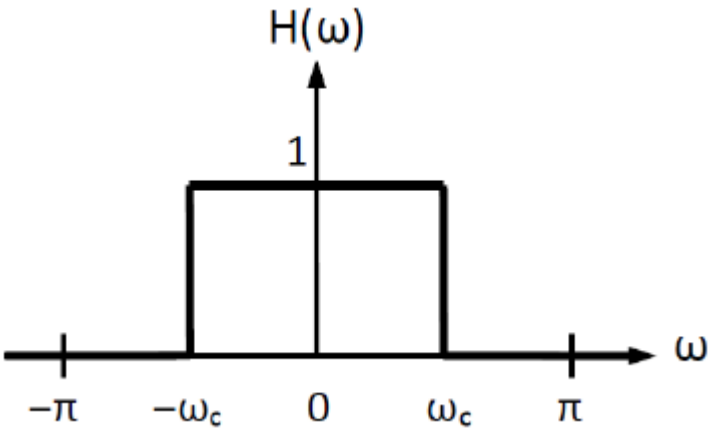
Chebyshev and FIR design

Practice questions · Day 20 · Thu 12 Nov · pack D12

5. Practice

From your course files

T9 Q1 (tutorial). Find the impulse response of the ideal lowpass filter with frequency response shown below. Explain why it is not possible to design an ideal lowpass filter in practice.



T9 Q1 figure (from the tutorial sheet)

T9 Q7 (tutorial). A digital lowpass filter is to designed by applying the bilinear transformation to an analogue Chebyshev type II filter. The digital filter is required to have the following specifications:

Passband ripple: ≤ 1 dB
Stopband attenuation: ≥ 50 dB
Passband edge frequency: 2 kHz
Stopband edge frequency: 4 kHz
Sampling frequency: 40 kHz

Determine ϵ and N for the analogue Chebyshev type II filter.

T9 Q8 (tutorial). Find the missing sample in the following impulse response to make it linear-phase:

$$h(n) = \{\underset{\uparrow}{1}, 2, *, -2, -1\}$$

Show that the resulting FIR system indeed has a linear phase response.

T9 Q9 (tutorial). A causal linear-phase FIR filter of length $M = 5$ is to be designed to approximate the ideal frequency response:

$$H_d(\omega) = \begin{cases} 1, & |\omega| < \pi/6 \\ 0, & \pi/6 \le |\omega| \le \pi. \end{cases}$$

(a) Determine the impulse response of the ideal filter.
(b) Obtain the FIR filter impulse response using the windowing method with a rectangular window.
(c) Repeat part (b) using the Hanning window.

T9 Q4 (tutorial; only if not done in D11). Given the following digital filter specifications, obtain the corresponding analogue filter specifications for bilinear transformation:

$$0.8 \le |H(\omega)| \le 1, \quad 0 \le \omega \le 0.3\pi$$

$$|H(\omega)| \le 0.1, \quad 0.6\pi \le \omega \le \pi.$$

T9 Q2 (tutorial; only if not done in D11). Convert the analogue filter

$$H(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9}$$

into a digital IIR filter using the bilinear transformation.

T9 Q3 (tutorial; only if not done in D11). An ideal analogue integrator is described by $H_a(s) = 1/s$.

(a) Convert the analogue integrator into a digital integrator using the bilinear transformation. Write the difference equation for the digital integrator.
(b) Roughly sketch the magnitude and phase response of the analogue integrator.
(c) Obtain the frequency response of the digital integrator $H(\omega)$, and roughly sketch the magnitude and phase responses.
(d) Compare the magnitude and phase responses in parts (b) and (c), and comment on how well they match.
(e) Considering that the digital integrator has a pole at $z = 1$, what restrictions may need to be placed on the input signal?

New questions

New Q1 [3 marks] An analogue Chebyshev Type I lowpass filter must have at most 0.5 dB passband ripple up to $\Omega_p = 1000$ rad/s, and at least 30 dB attenuation from $\Omega_s = 2000$ rad/s. (a) Find ϵ and N. (b) What order would a Butterworth filter need?

New Q2 [4 marks] Find the poles and the transfer function of a third-order Chebyshev Type I lowpass filter with 1 dB passband ripple and $\Omega_p = 1$ rad/s. At what passband frequencies is $|H| = 1$?

New Q3 [5 marks] A length-4 FIR filter has $h(n) = \{1, 3, *, *\}$ (origin at the first sample). (a) Give the two ways to complete $h(n)$ so that it has linear phase. (b) Which one can be used as a highpass filter? Justify using its zeros. (c) For that filter, write $H(\omega)$ in the form amplitude × linear phase, and state the group delay.

New Q4 [8 marks] A causal linear-phase FIR lowpass filter of length $M = 7$ is to approximate $H_d(\omega) = 1$ for $|\omega| < \pi/2$, 0 otherwise.

a. Find $h_d(n)$ and its values for $|n| \leq 3$. [2]
b. Find $h(n)$ with a rectangular window. [1]
c. Find $h(n)$ with a Hamming window. [2]
d. For both designs, compute $|H(0)|$, $|H(\pi/2)|$ and $|H(\pi)|$. Which window gives the better stopband? [2]
e. Name the linear-phase type and the group delay. [1]

New Q5 [5 marks] The digital lowpass prototype $H_{LP}(z) = \frac{1}{2}(1 + z^{-1})$ has its −3 dB edge at $\omega_p = \pi/2$. Use the L10 digital transformation table to make a highpass filter with passband edge $\omega_p' = \pi/3$. Find $H_{HP}(z)$ and check the gain at $\omega = \pi/3$ and $\omega = \pi$.

New Q6 [6 marks] A digital lowpass filter needs at most 0.5 dB ripple for $0 \leq \omega \leq 0.2\pi$ and at least 40 dB attenuation for $0.35\pi \leq \omega \leq \pi$. It will be designed by applying the bilinear transformation ($T = 1$) to an analogue prototype.

a. Find the pre-warped edges, and ϵ and N for a Chebyshev Type II prototype. [3]
b. How do the Butterworth and elliptic designs compare in order and phase? [1]
c. Using the course Type II response, find the analogue stopband frequencies where $|H| = 0$. [2]

6. Quick recall check

1. Which filter family has the lowest order for a given spec, and which has the best phase?
2. A Chebyshev I filter has $N = 4$ and 2 dB ripple. What is $|H(0)|$ in dB?
3. $h(n) = \{2, *, *, -1, -2\}$. Fill it in to make it linear phase.
4. What is the course Hamming window for $M = 3$?
5. Why can a Type II FIR filter (symmetric, even M) never be a highpass filter?

DSP · MOCK

DSP mock A

Practice questions · Day 22 · Sat 14 Nov · pack DX1

Instructions. Time allowed: **2 hours**. Open book: hard copy material is permitted. Attempt all questions. There are **60 marks** in total. Marks for each part are given in brackets. Show your working: method marks are given even when a number is wrong.

Budget about **2 minutes per mark**. Leave 5 minutes at the end to check units, ROCs and signs. If you get stuck on an early part, use the "assuming the answer to part (...) is ..." value and keep going.

This is a mock sat 4 days before the DSP exam. It covers L02–L10: about 25% on L02–L04 (Q1–Q2, 15 marks) and 75% on L05–L10 (Q3–Q8, 45 marks). Notation follows the course: F and Ω for continuous time, f and ω for discrete time, and $\operatorname{sinc} x = \sin x / x$.

Do not scroll past the horizontal rule until time is up.

Question	Topic	Lectures	Marks
1	Sampling, aliasing and system properties	L02–L03	8
2	LCCDE, impulse response and convolution	L04	7
3	Frequency response and DTFT properties	L05	7
4	DFT, circular convolution and resolution	L06	8
5	z-transform, ROC and inverse	L07–L08	8
6	Notch and comb filters	L09	7
7	Butterworth IIR design by bilinear transform	L10	9
8	Windowed linear-phase FIR design	L10	6
	Total		**60**

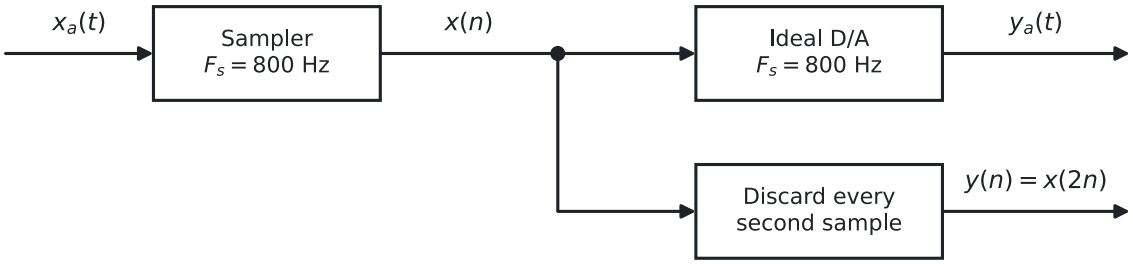
Question 1 [8 marks]

Sampling and system properties

The analogue signal

$$x_a(t) = 3\cos(300\pi t) + 2\sin(1100\pi t) \qquad (t \text{ in seconds})$$

is sampled at $F_s = 800$ Hz to give $x(n) = x_a(nT)$.



Question 1: the sampling chain, with the ideal D/A of part (c) and the sample-dropping system of part (d)

a. Determine the Nyquist rate for $x_a(t)$. Is $F_s = 800$ Hz adequate? **[1]**

b. Write $x(n)$ using only digital frequencies in the principal range $-\pi \le \omega \le \pi$. **[2]**

c. If $x(n)$ is passed through an ideal D/A converter at $F_s = 800$ Hz, what signal $y_a(t)$ is reconstructed? **[1]**

d. Every second sample of $x(n)$ is discarded, giving $y(n) = x(2n)$. Show that $y(n)$ is a single sinusoid. Write it as $A\cos(\omega_0 n + \phi)$ and give A, ω_0 and ϕ. **[2]**

e. For each system, state whether it is linear, time invariant, causal, static (memoryless) and BIBO stable. Give a one-line reason for each "no". **[2]**

f. $y(n) = x(n) - 0.5\,n\,x(n-1)$

ii. $y(n) = |x(n+1)|$

Question 2 [7 marks]

DT LTI systems

A causal DT LTI system is described by the LCCDE

$$y(n) - \tfrac{1}{4}y(n-1) - \tfrac{1}{8}y(n-2) = x(n) + x(n-1).$$

a. Find the characteristic modes of this system. Hence state whether it is BIBO stable. **[1]**

b. Using recursion, evaluate $y(n)$ for $0 \le n \le 3$ when the input is $x(n) = \left(\tfrac{1}{3}\right)^n u(n)$. Assume $y(n) = 0$ for $n < 0$. **[2]**

c. Find the impulse response $h(n)$. **[2]**

d. Assuming the answer to part (c) is $h(n) = \left[2\left(\tfrac{1}{2}\right)^n - \left(-\tfrac{1}{4}\right)^n\right]u(n)$, evaluate the convolution $y(n) = h(n) * x(n)$ in closed form, where $x(n) = \left(\tfrac{1}{3}\right)^n u(n)$. Check your answer against part (b) at $n = 0$ and $n = 1$. **[2]**

Question 3 [7 marks]

Frequency response

An FIR system is described by

$$y(n) = 0.5\,x(n) + x(n-1) + 0.5\,x(n-2).$$

a. Find the frequency response $H(\omega)$. Write it as a real amplitude times a linear-phase term, and give $|H(\omega)|$ and $\theta(\omega)$. **[2]**

b. Does this system have linear phase? State its group delay. **[1]**

c. Without computing any integral explicitly, evaluate $\displaystyle\int_{-\pi}^{\pi} |H(\omega)|^2\, d\omega$. **[1]**

d. Determine the steady-state output for the input **[3]**

$$x(n) = 4 + 2\cos\!\left(\tfrac{\pi}{2}n + \tfrac{\pi}{6}\right) + 3\cos\!\left(\tfrac{2\pi}{3}n\right) + 5(-1)^n.$$

Question 4 [8 marks]

The DFT

Consider the sequences

$$x_1(n) = \{\underset{\uparrow}{1}, 2, 0, -1\}, \qquad x_2(n) = \{\underset{\uparrow}{2}, 1, 1\}.$$

a. Compute the linear convolution $x_1(n) * x_2(n)$. **[1]**

b. Using your result from part (a), find the 4-point circular convolution $x_1(n) \circledast_4 x_2(n)$. What is the smallest DFT length N for which circular convolution gives the linear convolution? **[2]**

c. Compute the 4-point DFT $X_1(k)$ of $x_1(n)$. **[2]**

d. A signal contains two tones at 1.00 kHz and 1.03 kHz. It is sampled at $F_s = 8$ kHz and analysed with an $N = 2^m$-point DFT. Find the minimum N that gives a frequency resolution of 30 Hz or better, and the corresponding minimum record length T_0. **[2]**

e. A colleague records only $L = 200$ samples and zero-pads them to $N = 512$ before taking the DFT. Will the two tones be resolved? Explain in one or two sentences. **[1]**

Question 5 [8 marks]

z-transforms

An LTI system has transfer function

$$H(z) = \frac{3 + z^{-1}}{1 + 1.5z^{-1} - z^{-2}}.$$

a. Find the poles and zeros of $H(z)$. **[1]**

b. List all possible ROCs. For each ROC, find $h(n)$ using partial fractions. **[3]**

c. Which ROC gives a stable system? Is that stable system causal, anticausal or noncausal? Explain. **[2]**

d. Assuming the system is causal, write the difference equation that describes it. **[1]**

e. The stable system from part (c) is driven by $x(n) = \cos(\pi n)$ for all n. Find the output $y(n)$. **[1]**

Question 6 [7 marks]

Digital filters

a. A data-acquisition system samples at $F_s = 1$ kHz and picks up a 125 Hz interference tone. Design a second-order FIR notch filter

$$H(z) = b_0 \left(1 - 2\cos\omega_0\, z^{-1} + z^{-2}\right)$$

that removes 125 Hz completely and has $|H(\omega)| = 1$ at $\omega = \pi$. Give ω_0 and b_0. **[2]**

b. Find the gain $|H(\omega)|$ of your notch filter at 0 Hz and at 250 Hz. Comment briefly on what this says about the notch width. **[1]**

c. Does this notch filter have a stable causal inverse? Explain using its zeros. **[1]**

d. A different signal is sampled at $F_s = 1.2$ kHz and is corrupted by a 200 Hz hum and all of its harmonics. You decide to use the moving-average comb filter

$$y(n) = \frac{1}{M+1}\sum_{k=0}^{M} x(n-k).$$

Find the value of M that places nulls at 200 Hz and its harmonics. List every frequency in $0 < F \le F_s/2$ that is removed. **[2]**

e. What is the delay (in samples and in ms) that this comb filter introduces? What is its DC gain? **[1]**

Question 7 [9 marks]

IIR filter design

A digital lowpass filter is required to meet these specifications:

- Passband ripple: ≤ 3 dB
- Stopband attenuation: ≥ 15 dB
- Passband edge frequency: 1 kHz
- Stopband edge frequency: 2 kHz

- Sampling frequency: 8 kHz

The filter is to be designed by applying the bilinear transformation to an analogue Butterworth filter. Take $T = 1$.

a. Find the digital band edge frequencies ω_p and ω_s. **[1]**

b. Prewarp them to obtain the analogue edge frequencies Ω_p and Ω_s. **[1]**

c. Convert the specifications to linear scale and find the Butterworth order N. **[2]**

d. Find the cutoff frequency Ω_c so that the stopband specification is met exactly, and write the analogue transfer function $H(s)$. **[2]**

e. Assuming the answer to part (d) is $\Omega_c = 0.85$ rad/s, apply the bilinear transformation to obtain $H(z)$ with $a_0 = 1$. Write the difference equation of the digital filter, and check H at $\omega = 0$ and $\omega = \pi$. **[3]**

Question 8 [6 marks]

FIR filter design

A causal linear-phase FIR filter of length $M = 7$ is to approximate the ideal lowpass response

$$H_d(\omega) = \begin{cases} 1, & |\omega| < \pi/4 \\ 0, & \pi/4 \le |\omega| \le \pi. \end{cases}$$

a. Find the impulse response $h_d(n)$ of the ideal filter. **[1]**

b. Obtain the FIR impulse response $h(n)$, $0 \le n \le 6$, by the windowing method with a **Hamming** window. Give the values to 4 decimal places. **[3]**

c. Is $h(n)$ symmetric or antisymmetric? State the group delay of the filter. **[1]**

d. Compute $H(0)$. Explain why it is not 1, and say how you would make the DC gain exactly 1. **[1]**

End of questions

DSP · MOCK

DSP mock B

Practice questions · Day 24 · Mon 16 Nov · pack DX2

Instructions

- Time allowed: **2 hours**. Sit it in one go, timed, with your printed notes, Test 1 Reference and reference sheet beside you (open book, hard copy only).
- Attempt **all 11 questions**. There are **60 marks** in total. Marks for each part are given in brackets. Budget about 2 minutes per mark, and keep the last 5 minutes to check.
- Show your working. Most marks go to method, so a wrong number with a correct method still scores.
- Some parts say "assuming the answer to part (x) is …". Use the value given there, even if your own answer is different. This stops one slip from costing you the rest of the question.
- Notation is the course's. ω is DT frequency (rad/sample), Ω is CT frequency (rad/s). $\operatorname{sinc} x = \sin x / x$. The arrow $\uparrow$ marks the $n = 0$ sample. Use $T = 1$ for every bilinear-transform calculation.
- This paper is meant to feel different from mock A. It uses the less common angles of L03–L10. Do not open the marking guide until time is up.

Question	Topic	Lectures	Marks
1	Correlation and delay estimation	L03	5
2	LCCDE with complex modes	L04, L05	6
3	DTFT shortcut evaluations	L05	5
4	Deconvolution via the DFT	L06	6
5	Overlap-save block convolution	L06	6
6	Two-sided inverse z-transform: every ROC	L07, L08	6
7	Unilateral z-transform with initial conditions	L08	6
8	Digital resonator and allpass filter	L09	6
9	Chebyshev type I order and ϵ	L10	5
10	Completing a linear-phase FIR filter	L10	5
11	Short answers	L06, L10	4
		Total	**60**

Question 1 [5 marks]: Correlation and delay estimation

A short radar pulse $x(n) = \{\underset{\uparrow}{1}, 2, -1\}$ is transmitted. The ADC samples the received echo at $F_s = 10$ MHz:

$$y(n) = \{\underset{\uparrow}{1}, 0, -1, 2, 4, -2, 1\}, \qquad y(n) = 0 \text{ elsewhere.}$$

Use the lecture definitions $r_{xx}(l) = \sum_n x(n)x(n-l)$ and $r_{yx}(l) = \sum_n y(n)x(n-l)$.

a. Compute the autocorrelation $r_{xx}(l)$ for all l. **[1]**

b. Compute $r_{yx}(l)$ for $-2 \le l \le 6$. **[2]**

c. Estimate the delay $\hat{d}$ and the target range r. Take $c = 3 \times 10^8$ m/s. **[1]**

d. Model the echo as $y(n) \approx a\, x(n - \hat{d})$ plus interference. Use your results from (a) and (b) to estimate the gain a. State the range resolution of this radar (the change in range for a one-sample change in delay). **[1]**

Question 2 [6 marks]: LCCDE with complex modes

A causal DT LTI system is described by

$$y(n) - y(n-1) + 0.5\, y(n-2) = x(n).$$

a. Find the characteristic modes in polar form. State whether the system is stable, and give your reason. **[1]**

b. Using recursion, evaluate $h(n)$ for $0 \le n \le 6$. Assume the system is initially at rest. **[1]**

c. Find a closed-form, real-valued expression for the impulse response $h(n)$. **[2]**

d. Find the steady-state output when the input is $x(n) = 4 + 2\cos\left(\frac{\pi}{2}n\right)$ for all n. **[2]**

Question 3 [5 marks]: DTFT shortcut evaluations

Consider the signal

$$x(n) = \{1, -2, \underset{\uparrow}{3}, 0, -1, -4\}$$

with DTFT $X(\omega)$. Compute each quantity below **without** finding $X(\omega)$ explicitly.

a. $X(0)$ and $\angle X(0)$. **[1]**

b. $X(\pi)$. **[1]**

c. $\displaystyle\int_{-\pi}^{\pi} X(\omega)\, e^{j2\omega}\, d\omega$. **[1]**

d. $\int_{-\pi}^{\pi} |X(\omega)|^2 \, d\omega$. **[1]**

e. $\int_{-\pi}^{\pi} \left| \frac{dX(\omega)}{d\omega} \right|^2 d\omega$. **[1]**

Question 4 [6 marks]: Deconvolution via the DFT

An unknown input $x(n)$ of length 3 passes through a known FIR channel $h(n) = \{\underset{\uparrow}{2}, 1\}$. The complete output is recorded:

$$y(n) = \{\underset{\uparrow}{4}, 0, 1, 1\}.$$

a. Explain why a 4-point DFT is enough to recover $x(n)$ exactly from $y(n)$, with no time-aliasing. **[1]**

b. Compute the 4-point DFTs $Y(k)$ and $H(k)$. **[2]**

c. Hence find $X(k)$, and use the 4-point IDFT to recover $x(n)$. **[2]**

d. Suppose the channel were instead $h(n) = \{\underset{\uparrow}{1}, 1\}$. Explain why this method then fails with a 4-point DFT. **[1]**

Question 5 [6 marks]: Overlap-save block convolution

An FIR filter $h(n) = \{\underset{\uparrow}{1}, -1, 2\}$ filters the input

$$x(n) = \{\underset{\uparrow}{1}, 2, 0, -1, 3, 1\}.$$

You must use the **overlap-save** method with 4-point circular convolutions ($N = 4$).

a. State the number of new input samples per block, L, and how many output samples of each block are discarded. **[1]**

b. Write down every 4-sample input block you need to produce the full linear-convolution output. **[1]**

c. Compute the 4-point circular convolution of each block with $h(n)$ (you may compute only the samples you keep). Assemble $y(n) = h(n) * x(n)$. **[3]**

d. For the first block, compute the two discarded samples. Explain why they differ from the true $y(-2)$ and $y(-1)$. **[1]**

Question 6 [6 marks]: Every possible sequence for each ROC

$$X(z) = \frac{1 + 11z^{-1} - 11z^{-2}}{(1 - 0.5z^{-1})(1 - 2z^{-1})(1 + 3z^{-1})}$$

a. Expand $X(z)$ in partial fractions of the form $\dfrac{A}{1 - p\,z^{-1}}$. **[2]**

b. Assume the answer to part (a) is

$$X(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{1}{1 - 2z^{-1}} - \frac{1}{1 + 3z^{-1}}.$$

List every possible ROC. For each one, give the sequence $x(n)$ and classify it as causal, anticausal or two-sided. **[3]**

c. Using the expression in (b), which ROC gives a sequence that has a DTFT? For that sequence, evaluate $x(0)$ and $x(-1)$. **[1]**

Question 7 [6 marks]: Unilateral z-transform with initial conditions

A system is described by

$$y(n) - 0.75\,y(n-1) + 0.125\,y(n-2) = x(n),$$

with input $x(n) = 3u(n)$ and initial conditions $y(-1) = 2$, $y(-2) = 4$.

a. Take the unilateral z-transform of both sides, including the initial conditions, and obtain $Y^{+}(z)$. **[2]**

b. Find $y(n)$ for $n \geq 0$. **[3]**

c. Split your answer into its zero-input and zero-state parts. Check $y(0)$ and $y(1)$ by recursion. **[1]**

Question 8 [6 marks]: Digital resonator and allpass filter

A digital resonator is to pick out a 1 kHz tone from a signal sampled at $F_s = 8$ kHz. Use the course's resonator form, with poles at $re^{\pm j\omega_0}$, $r = 0.9$, and zeros at $z = \pm 1$.

a. Write $H(z)$ with numerical denominator coefficients. State which input frequencies (in Hz) it rejects completely. **[1]**

b. Choose b_0 so that $|H(\omega_0)| = 1$, and write the difference equation. **[2]**

A first-order allpass filter in the course form $H_{ap}(z) = z^{-N}A(z^{-1})/A(z)$ has $A(z) = 1 - 0.8z^{-1}$.

c. Write $H_{ap}(z)$. Give its pole and zero, and show that $|H_{ap}(\omega)| = 1$. **[1]**

d. Find the phase $\angle H_{ap}(\omega)$ at $\omega = \pi/2$. **[1]**

e. Find the group delay at $\omega = 0$. Explain in one line why such a filter is useful. **[1]**

Question 9 [5 marks]: Chebyshev type I order and ϵ

A digital lowpass filter will be designed by applying the bilinear transformation ($T = 1$) to an analogue Chebyshev type I filter. The specifications are:

- passband ripple ≤ 0.5 dB, passband edge 1 kHz
- stopband attenuation ≥ 40 dB, stopband edge 2.5 kHz
- sampling frequency 10 kHz

a. Find ω_p and ω_s, and the prewarped analogue edges Ω_p and Ω_s. **[1]**

b. Find δ_p, δ_s and ϵ. **[1]**

c. Find the filter order N. **[2]**

d. Assume the answer to part (c) is $N = 5$. Find the attenuation (in dB) that the analogue filter actually gives at Ω_s. **[1]**

Question 10 [5 marks]: Completing a linear-phase FIR filter

$$h(n) = \{-1, \underset{\uparrow}{2}, 4, *, -2, 1\}$$

a. Find the missing sample $h(3)$ that makes this filter linear-phase. Give your reason. **[1]**

b. Write $H(\omega)$ in the form $A(\omega)\, e^{j\theta(\omega)}$, with $A(\omega)$ real. **[2]**

c. Give the phase response and the group delay. **[1]**

d. Evaluate $H(0)$ and $H(\pi)$. Could this filter be used as a lowpass filter? Explain. **[1]**

Question 11 [4 marks]: Short answers

a. Give two reasons why the ideal lowpass filter cannot be realised in practice. **[1]**

b. A signal holds two tones at 1000 Hz and 1050 Hz. It is sampled at $F_s = 8$ kHz. A student records $L = 64$ samples and zero-pads them to $N = 1024$ before taking the DFT. State the DFT bin spacing. Can the two tones be resolved? What is the minimum record length L (and duration T_0) needed? **[2]**

c. In Question 9, a student forgets to prewarp and uses $\Omega_s = \omega_s / T = \pi/2$ rad/s directly. After the bilinear transformation, at what frequency (in Hz) does the stopband edge actually land? What does this show about the bilinear map? **[1]**

End of questions